



# Meen Patel

## Education

**Nirma University**

*B.Tech in ECE*

**2021 – 2025**

*Ahmedabad, Gujarat*

**Vishwabharati English Medium School**

*12<sup>th</sup>*

**2020 – 2021**

*Ahmedabad, Gujarat*

## Experience

**NXON AI Pvt. Ltd.** | *Microcontrollers, espressif*

*Embedded Systems Engineer*

*Embedded Software Developer Intern*

**Dec. 2024 – Present**

*April 2025 – Present*

*Dec. 2024 – April 2025*

- Assigned key responsibilities included customizing and enhancing open-source firmware based on specific client requirements, encompassing user interaction features and system performance optimization.
- Developed a Gemini-powered project on ESP32 with speaker and microphone integration.
- Worked on vision module for drone along with a senior engineer

**eInfochips** | *Research*

**Jun. 2024 – Jul. 2024**

- I explored the applications of AI in electronics and researched a company that manufactures electronic products and provides solutions for wireless IoT connectivity.

**Midas Metcons Pvt. Ltd.** | *Embedded C, 8051*

**Jun. 2023 – Dec. 2023**

- I worked on technologies for automated storage systems and programmed it using microcontrollers and simulating techniques

## Skills

**KEY EXPERTISE:** Embedded C, Embedded Systems, Circuit Design, Electronic Circuit Design, Image Processing, Assembly Language, Python, Arduino, MASM, ModelSim, MATLAB, Proteus, Firmware, Digital Electronics, 8051 Microcontroller, C, Keil, Automation, Industrial Automation, IoT, Communication Protocols.

## Projects

**ESP32 AI Voice Assistant**

**Mar 2024 – May 2024**

- Voice assistant using ESP32 and cloud APIs (Deepgram, Gemini).
- Features: Real-time STT, NLP response, TTS feedback.

**Vertical Lift Module (Warehouse Automation)**

**Mar 2024 – May 2024**

- Built control system on ESP32 with RTOS for improved task management and efficiency.
- Implemented IoT remote monitoring system.

**MIPS Assembler in Python**

**Mar 2024 – Apr 2024**

- Implemented a script that translates assembly instructions into machine code with error handling to demonstrate assembler functionality.

**Image Modulation Using Sound (MATLAB)**

**Sept 2023 – Nov 2023**

- Dynamically changed image based on sound frequency and amplitude.

**Smart Irrigation with 8051**

**Mar 2023 – May 2023**

- Designed digital irrigation using 8051 and Proteus simulation.

## Achievements

**AMD Pervasive AI Developer Contest**

**Feb 2024 – Aug 2024**

*Neuro Glove: Smart Rehabilitation Assistant*

*Hackster.io*

**Smart India Hackathon**

**Sep 2023**

*Regional Level Finalist*

*Nirma University*

## About

I am an Electronics and Communication Engineering graduate with a strong interest in core electronics, particularly in embedded system design. I have gained hands-on experience and am working on enhancing my skills through practical learning and real-world applications.

## Contact

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